

ECE448 Spring 2025

Lesson 29

ADS-B



UNITED STATES
AIR FORCE
ACADEMY

NOMINAL SCHEDULE

Lesson	Topics	Notes
22	RF Reverse Engineering	Final Project Proposal due
23	RF Reverse Engineering	
24	RF Reverse Engineering	
25	Flex day	RF RE Project Due
26	ADS-B	
27	Project Work Day	Background Research due
28	ADS-B	
29	ADS-B	
30	AIS	
31	AIS	Design Description due
32	WBFM Radio	ADS-B/AIS Project
33	HD Radio	
34	HD Radio	
35	Project Work Day	Prototype Description due
36	QAM	
37	QAM /OFDM	
38	QAM /OFDM	
39	Project Day	
40	Final Project Presentations	Final Project due

LESSON 29 LEARNING OBJECTIVES

- 1.1 Articulate the purpose and implementation of ADS-B and Mode S
- 1.2 Understand the encoding and modulation schemes of ADS-B Mode S messages
- 1.3 Use existing tools to receive and decode ADS-B messages
- 1.4 Develop a PPM modulation block

ADS-B

- Automatic Dependent Surveillance Broadcast (ADS-B)
- Mode S (Mode Select Beacon System) is a broader standard eventually adopted as ADS-B implementation
- Mode S includes many frameworks for uplink/downlink
- ADS-B most typically refers to **Mode S Extended Squitter (ES)**, which use Downlink Format (DF) 17 messages.

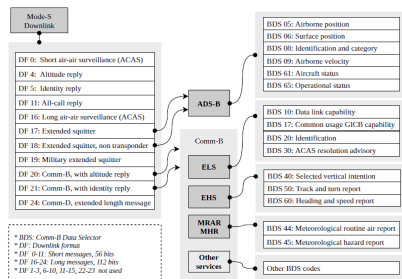
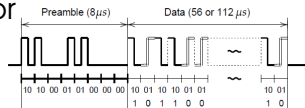


Figure: Source: Sun, Junzi. "The 1090 Megahertz Riddle: A Guide to Decoding Mode S and ADS-B Signals."

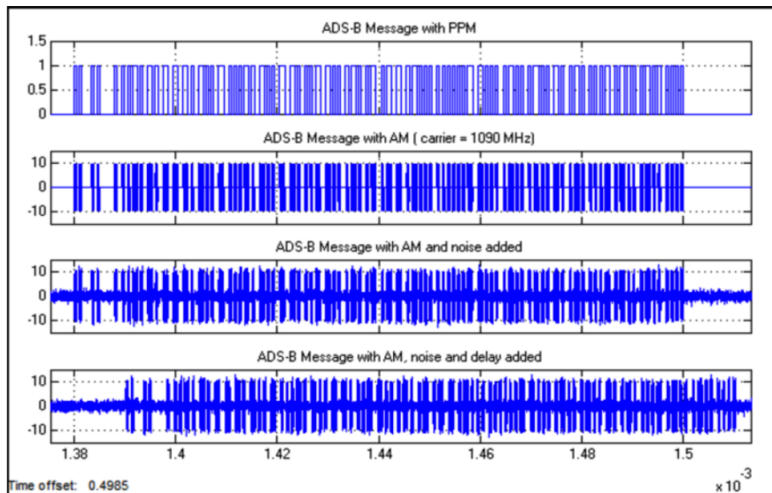
ADS-B PHYSICAL LAYER

- ADS-B is broadcast on 1090 MHz or 978 MHz

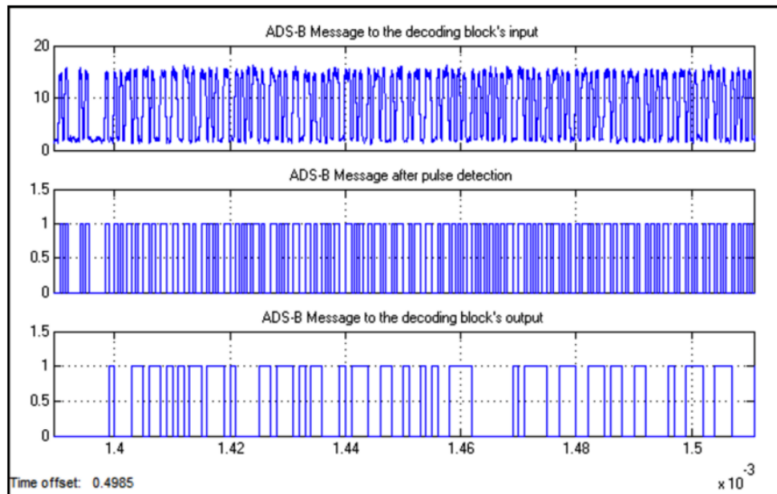


- ADS-B downlink uses PPM (AM) modulation
- [Reference Info](#)
- Each bit is $1\mu\text{s}$ long
- A "1" bit is represented by $0.5\mu\text{s}$ pulse and $0.5\mu\text{s}$ "quiet" time
- A "0" bit is represented by $0.5\mu\text{s}$ quiet time and $0.5\mu\text{s}$ pulse

ADS-B OUT GENERATOR



ADS-B OUT GENERATOR



ADS-B MESSAGE STRUCTURE

DF (5)	CA (3)	ICAO (24)	ME (56)	PI (24)
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Bit	No. bits	Abbreviation	Information
1 -5	5	DF	Downlink Format
6-8	3	CA	Transponder Capability
9-32	24	ICAO	ICAO Aircraft address
33-88	56	ME	Message, extended squitter
89-112	24	PI	Parity/Interrogator ID

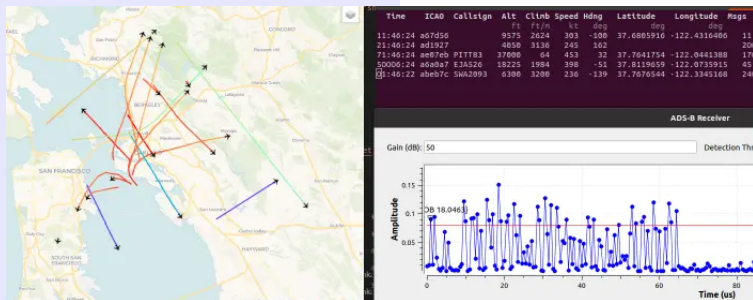
EXAMPLE ADS-B "SENTENCE"

8D4840D6202CC371C32CE0576098

HEX	8D	4840D6	202CC371C32CE0	576098
BIN	10001 101	010010000100	[00100]0000010110011	010101110110
		000011010110	00001101110001110000	000010011000
			110010110011100000	
DEC	17 5		[4]	
	DF CA	ICAO	ME	PI
			[TC]	

TOOLS

- [dump1090](#)
- SDR Angel
- [gr-adsb](#)



REFERENCES

1. Sun, Junzi. [The 1090 Megahertz Riddle: A Guide to Decoding Mode S and ADS-B Signals](#). 2021. TU Delft OPEN Publishing.
2. [DRWXR ADS-B Decoding](#)
3. [HABR ADS-B Demod and Decode](#)

REFERENCES

1. Sun, Junzi. [The 1090 Megahertz Riddle: A Guide to Decoding Mode S and ADS-B Signals](#). 2021. TU Delft OPEN Publishing.
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